

Capacity-Constrained Clustering for Auditable Redistricting Construction

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Abstract

This paper introduces T.15, a capacity-constrained clustering construction family for BISECT. The implementation emphasizes deterministic seeding, population-capacity status, repair-aware summaries, and RPLAN audit sidecars. The paper frames clustering as a construction alternative with explicit status and repair evidence, not as a claim of global optimality, legal compliance, community preservation, or political fairness.

1 Introduction

Capacity-constrained clustering gives BISECT a construction family whose primary abstraction is assignment to population-capacity clusters. This makes capacity status, repair, and infeasibility visible rather than hidden inside a single final assignment.

The central design choice is to make failure part of the artifact contract. Many construction algorithms only expose a final plan or an error. T.15 exposes the intermediate accountability surface: which deterministic seeds were chosen, which assignment rule was used, whether capacity was satisfied, whether repair was needed, and whether the exported plan is backed by RPLAN and RCTX sidecars. That makes the paper useful even before broad empirical comparison, because it defines how a clustered construction should explain itself.

2 Method

The staged method uses deterministic farthest-point seeds, assigns units subject to population capacity, and records whether the resulting assignment is **valid**, **needs-repair**, or **infeasible-capacity**. Small repair routines are treated as auditable post-processing rather than invisible cleanup.

The atlas view makes the key decision inspectable: a geographically nearest cluster is only a candidate, not an entitlement. If assigning a unit would overfill that cluster, the candidate is rejected with visible capacity arithmetic and the next eligible cluster is tried. The paper’s artifact claim therefore depends on preserving seed order, rejected capacity candidates, fallback assignments, and final status.

2.1 Reproducible Procedure

The constructor operates on the graph and unit attributes supplied by the BISECT state pipeline. The current slice is graph-based and capacity-oriented: community-character weights and richer attribute clustering are future or optional inputs rather than claims of this baseline.

1. Select deterministic farthest-point seeds from the available units, breaking ties by stable unit identifier.
2. Assign remaining units to candidate clusters in deterministic order while respecting target population capacity and declared slack.
3. Mark the assignment **valid** when population deviation and connectedness checks pass.
4. Mark the assignment **needs-repair** when the assignment is close enough to repair but does not yet pass all checks.
5. Run bounded repair only for **needs-repair** outputs. In the current slice this is an exhaustive small-fixture repair path.
6. Mark the result **infeasible-capacity** when the requested capacity profile cannot be satisfied. Invalid input is returned as an error before a plan artifact is emitted.

Status	Meaning	Artifact behavior
valid	Assignment satisfies declared profile	Write final sidecars with repair-status=not-needed or repaired
needs-repair	Assignment failed connectedness or balance checks but may be repairable	Keep repair requirement visible; do not collapse into final export
infeasible-capacity	Capacity cannot be satisfied under the declared tolerance	Write witness summary rather than a final plan claim
invalid input	Input shape is malformed, such as $k = 0$ or mismatched arrays	Return an error before producing a plan artifact

Table 1: Capacity-clustering status taxonomy. Repair is visible lineage, not hidden cleanup.

2.2 Worked Capacity Decision

The assignment step is easiest to read as a visible rejection table. Suppose two clusters target 50 population each with ten percent slack, so the local upper capacity is 55. The next unit has population 20. The nearest seed is not automatically allowed to take it; capacity arithmetic decides which candidate remains eligible.

Candidate cluster	Distance rank	Current pop.	Pop. after unit	Decision
A	1	45	65	reject: exceeds 55
B	2	32	52	accept: within slack

Table 2: T.15 assigns by deterministic ranked candidates subject to capacity. The nearest cluster can lose when accepting the unit would overfill it.

3 Implementation

The implementation lives in `bisect-clustering` and is exposed through the capacity-clustering structure mode in the CLI and YAML configuration. Summaries record capacity status, repair status, parameter hashes, and algorithm lineage.

The crate boundary is split across deterministic seed placement, capacity-aware assignment, small-fixture repair, metrics, and output summaries. The summary schema records the method, seed method, repair method, capacity status, repair status, population deviation, edge cut, and a deterministic parameter hash. The CLI layer uses the repaired capacity-clustering entry point for exported construction paths and rejects non-valid results before final plan packaging.

4 Audit Artifacts

Capacity-clustering outputs are framed as final-plan candidates only after the RPLAN and RCTX audit sidecars record assignments, capacity status, repair status, parameter hashes, and algorithm lineage. The verifier checks the declared audit profile; it does not certify that clustering choices satisfy fairness or legal requirements beyond the profile.

The sidecar contract binds the exported assignment to the constructor family, declared parameters, capacity and repair status, validation result, and algorithm lineage. Parameter hashes are used to detect drift between the reported run configuration and the artifact under verification. If repair changes assignments, that status remains part of the lineage rather than being collapsed into an ordinary valid plan.

The audit fixed point is therefore narrower than the construction method: it binds the emitted plan to a recorded clustering run and to the declared legal profile. It does not certify that the chosen seeds respect communities of interest, that a repaired assignment is normatively preferable to another valid assignment, or that the capacity rule captures state-specific subdivision-preservation law.

5 Evaluation Plan

The current evidence layer includes deterministic seed tests, infeasible capacity fixtures, connectedness checks, RPLAN audit package tests, and a benchmark-tier RPLAN package generated from the capacity constructor on a 100-unit synthetic path. The next empirical layer should measure feasibility and quality on selected real-data smoke states.

Claim	Evidence now	Missing evidence	Status
Deterministic seeds	L0 seed fixtures	None for fixtures	Current
Capacity status reporting	L0 infeasibility tests	Broader case sweep	Current
RPLAN repair lineage	L1 sidecar tests	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke
Plan quality vs. baselines	None in this paper	Metrics and datasets	Future

Table 3: Evidence ladder for T.15 claims. Feasibility and quality are tracked separately.

Constructor	Capacity behavior	Failure artifact	Current role
Capacity clustering	explicit status/repair	witness or repair lineage	baseline
Spectral	balance via sweep	validation failure	comparison
Regionalization	capacity during merges	merge/status summary	comparison
Flow construction	capacity flow status	infeasibility witness	comparison
METIS	external partitioning	manifest/validation result	re-comparison

Table 4: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

The comparison target is not just score quality. T.15 should be judged by whether it explains capacity pressure better than alternatives: which units could not be assigned, whether repair changed the plan, and what verifier evidence exists after export. Quality metrics such as compactness, subdivision splits, and partisan measures remain future empirical comparisons until the same states and tolerances are run across the construction portfolio.

6 Limitations

The current slice is not a full clustering benchmark suite. It establishes the artifact contract and deterministic baseline first; broader claims about quality, compactness, or communities of interest require additional sweeps.

Capacity feasibility is also not a normative districting claim. A valid or repaired output can still be weak on compactness, community preservation, partisan symmetry, or legal criteria outside the declared audit profile. Community-character weights are treated as future or optional evidence, not as part of the baseline claim.

The repair path is deliberately bounded. Exhaustive small-fixture repair is excellent for reproducibility tests, but it is not evidence that large real states can always be repaired or that repair will preserve qualitative communities. Real-data claims require recorded commands, package outputs, and failure cases, not just successful synthetic fixtures.

7 Conclusion

T.15 adds clustering as a first-class construction family while preserving the audit fixed point: final plans must converge into RPLAN, RCTX, audit certificates, and manifests. The contribution is the inspectable capacity and repair contract, not a claim that clustered plans are globally optimal, politically fair, or legally sufficient.

That contract makes T.15 a useful companion to spectral, regionalization, and flow construction. It gives future comparison papers a baseline that can say not only what plan was produced, but also whether capacity pressure caused repair, infeasibility, or a clean final export.

References

BISECT Project. T.15 capacity-constrained clustering implementation spec, 2026. BISECT internal specification.